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An efficient late-interaction ModernBERT architecture for interpretable token-level semantic matching in Turkish

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ABSTRACT

Developing models that balance contextual understanding and computational efficiency remains a challenge in natural language processing, particularly for morphologically rich languages such as Turkish. This study proposes a hybrid Late-Interaction Fusion model based on ModernBERT to improve token-level semantic matching in Turkish texts. The model preserves the scalability of Bi-Encoders while incorporating contextual depth through a Multi-Head Attention-based late-interaction mechanism. The novelty of this work is that, unlike ColBERT-style late-interaction retrievers which combine pre-computed representations through a parameter-free similarity operator, the proposed fusion layer is a learned, query-conditioned cross-attention block that emits a calibrated relevance probability for every individual token, and that it is instantiated on a Turkish ModernBERT backbone trained on a two-million-pair corpus built through evidence-grounded LLM-assisted annotation. Experimental results show that the model attains a token-level F1-score of 0.557, closely approaching the span-based GLiNER baseline (0.580) while requiring substantially less computation, and clearly outperforming lexical matching (0.065), dense retrieval (0.129) and a standard Bi-Encoder (0.481). Additionally, the shared encoder enables fast convergence and the highest Macro-F1 (0.597) among the compared Turkish BERT variants in a downstream classification task, even with limited training data.

Introduction

With the advancement of information technology, access to information has become one of the most important requirements of both individual and corporate processes [1,2]. From corporate search engines to customer service automation, and from academic research to legal document analysis, users across many fields demand quick access to accurate information within vast textual archives. Scenarios such as reviewing legal documents, generating quick responses in customer support systems, scanning text for academic research, or gaining insights from product reviews are directly linked to the automation capabilities provided by Natural Language Processing (NLP) techniques [3,4]. Traditional information retrieval systems mostly operate at the passage or document level. That is, while they are successful in determining what a document is generally about, they often fall short in the task of "marking" the few sentences or phrases that contain the actual answer the user needs [5].

In this context, "semantic token-level matching" stands out as an important research topic. When a query is entered, determining whether each word or sub-unit (token) within the text is related to the query not only indicates which document is relevant, but also makes it possible to pinpoint exactly where the information the user is looking for is located [6,7]. Thus, the process of accessing information becomes both more precise and more efficient in terms of user experience. However, in agglutinative and morphologically rich languages such as Turkish, this task is much more complex than in analytic languages such as English [8,9].

Various approaches have been developed in the literature to address these issues. Bi-Encoder architectures provide high efficiency and scalability by encoding queries and text independently of each other [10]. Additionally, establishing a relationship between the query and the text solely through vector similarity often results in missing deeper semantic connections. On the other hand, Cross-Encoder architectures process the query and text together, ensuring dense cross-attention between each token and generally achieving the highest performance [11]. However, this high accuracy is impractical for large-scale and real-time applications due to the significant increase in computational cost. Late interaction approaches (e.g., ColBERT and ColBERTv2), which aim to strike a balance between these two extremes, combine both efficiency and accuracy goals by balancing these two extremes [12,13].

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With the proliferation of Large Language Models (LLMs), data enrichment and zero-shot learning paradigms have also opened new horizons in this field [14,15]. Thanks to the generative capabilities of LLM models, the necessity of manually preparing large amounts of labeled data has been partially eliminated, making it possible to develop effective models in semi-supervised or zero-shot scenarios. This approach is particularly important for languages with limited resources [16,17]. Due to its morphological structure and limited open-source data sets, the Turkish language is considered one of the critical languages where these methods can be applied [18].

This study focuses on the problem of automatically detecting and highlighting query-related parts in Turkish texts at the token level, targeting the aforementioned gaps in the literature. In this study, a "Late-Interaction Fusion" architecture built on a ModernBERT-based model [19] is proposed [20]. Thus, the efficiency of the Bi-Encoder and the accuracy power of the Cross-Encoder are combined in a hybrid structure. Furthermore, a Focal Loss-based training strategy has been adopted to overcome the class imbalance problem [21], and LLM-based semi-supervised labeling has been used in the data enrichment process. The results obtained show that the proposed approach can compete with strong models in the literature in terms of both accuracy and efficiency.

This study aims to contribute to the field of NLP both theoretically and practically. In this context, the main contributions of the study can be summarized as follows:

- A Late-Interaction Fusion layer has been developed that combines the efficiency of Bi-Encoder with the accuracy of Cross-Encoder. This hybrid system provides both accuracy and efficiency advantages.
- A 500,000-word base corpus has been enriched to 2 million examples using an LLM-supported labeling strategy, providing a significant resource to the literature. This dataset is not only the foundation of the study but also an open contribution to the Turkish NLP community.
- An original platform called Zero-Shot Comparative Annotation Tool (ZSCAT) has been developed for use in the study; for data labeling, comparing model outputs, and human–LLM interactive verification tasks. This platform accelerates semi-supervised data production processes and contributes not only a model but also a practical tool to the literature.
- The proposed model was compared with classical approaches and powerful state-of-the-art models such as GLiNER [22], achieving competitive results in terms of both accuracy and efficiency.
- The method presented in this study offers a scalable roadmap not only for the Turkish language but also for other languages with similar morphological challenges [23]. In this regard, it is expected to make a valuable contribution to the future development of "token-level information access" and "text annotation" systems.

The remaining sections of this study are structured as follows: In the second section, related studies with similar characteristics to this research and published in reputable journals are discussed in detail. In the third section, the theoretical foundations and application details of the proposed method are presented. The fourth chapter contains the experimental findings and evaluations related to these findings. The fifth and final chapter summarizes the overall results of the study and shares projections for future research directions.

Related work

Research in the fields of Information Retrieval (IR) and Natural Language Processing (NLP) has led to significant advances, particularly in identifying and labeling relevant segments within large text corpora. This research can generally be examined along three main axes. In this context, the literature review has been conducted around three main topics: "classical deep learning-based text representations", "interaction-based architectures", and "zero-shot and multi-task learning approaches", which have gained prominence in recent years.

Traditional deep learning-based text representation approaches

Parikh et al. [24] achieved significant progress by incorporating attention mechanisms into classical deep learning representations with their "Decomposable Attention Model" developed for Natural Language Inference (NLI) tasks. The model encodes input sentences separately and then models their interaction by performing a soft-alignment operation between them. Although this structure is used for sentence-level prediction, token-level similarity calculations are performed in the intermediate layers. In experiments conducted on the SNLI dataset, the model achieved 86.3% accuracy and 84.2% F1-score. The model's low computational cost and high performance are noteworthy, especially when compared to independent encoder-based architectures.

Yin et al. [25] presented a classical approach to sentence pair modeling problems using an attention-based convolutional neural network architecture they named ABCNN. ABCNN combines three different types of attention (input attention, feature attention, and output attention) with CNN to enable multi-layered interaction between text pairs. The model achieved 81.1% precision, 79.5% recall, and 80.3% F1-score on the MSRP dataset. It was particularly successful in short sentence matching and in establishing semantic relationships beyond surface-level similarities.

Wang et al. [26] developed an approach that interacts across sentences using multiple views with the Bilateral Multi-Perspective Matching (BiMPM) model. BiMPM performs comprehensive matching at the token level by calculating multiple similarity scores for each word with its corresponding words in the opposing sentence. In experiments conducted on the SNLI and Quora Question Pairs datasets, the model achieved 88.5% accuracy and 85.3% F1-score on SNLI, and 89.0% precision, 86.7% recall, and 87.8% F1-score on Quora. These results demonstrate the strengths of word-based interactive architectures.

Chen et al. [27] achieved significant success in NLI tasks by considering word-level interactive information flow using the Enhanced Sequential Inference Model (ESIM), a classic LSTM-based approach. The model's basic structure involves applying cross-attention to input sentences encoded with BiLSTM to create word-level interactions. It has been one of the highest-performing models in the pre-BERT era, achieving 88.0% accuracy and 85.9% F1-score on the SNLI dataset.

Interaction-based architectures

The Bidirectional Attention Flow (BiDAF) model developed by Seo et al. [28] was developed specifically for question-answering systems and establishes deep interaction through bidirectional (context-to-query and query-to-context) attention calculations between text and questions. The model's greatest contribution is that it provides detailed word-level interaction by calculating the importance of each token in context from both the input (context) and query perspectives. In experiments conducted on the SQuAD dataset, BiDAF achieved a 77.3% F1-score and a 67.7% exact match (EM) score. It is an important reference point for tasks requiring token-level information extraction.

Tay et al. [29] proposed calculating deep semantic similarity between sentence pairs by using multiple attention mechanisms in parallel to increase the interaction level with the Multi-Cast Attention Network (MCAN) model. MCAN models both local and global interactions by incorporating attention components at various levels, such as self-attention, alignment-attention, and fusion-attention. Experiments conducted on the TrecQA dataset yielded 84.2% precision, 83.5% recall, and 83.8% F1-score. This architecture has been inspiring, especially for approaches seeking to perform multi-faceted comparisons at the token level.

Gong et al. [30] proposed a word-level multi-layer interaction model for NLI tasks in their study titled Inference-over-Interaction Network (IoI). In the model, the relationship between each word pair is deepened layer by layer using iterative attention mechanisms, and interaction with context is modeled simultaneously. On the MultiNLI dataset, the model achieved the best results of its time with 88.7% accuracy and 86.9% F1-score. Layered interaction strategies are particularly noteworthy for their applicability to morphologically rich languages such as Turkish.

Guo et al. [31] modeled the interaction between query and document words using a histogram-based approach in ad-hoc information retrieval tasks with the Deep Relevance Matching Model (DRMM). The similarity between each word pair was quantified using fixed-size bins and deepened in an input-interaction-output structure. They achieved 76.1% precision and 74.9% F1-score on the TREC Robust04 dataset. Although it is a retrieval-based structure, it is one of the pioneering studies that contributes to token-level semantic matching.

Zero-shot and multi-task learning approaches

Sanh et al. [32], in their study titled T0: Multitask Prompted Training Enables Zero-Shot Task Generalization, aimed to enable the model to perform successfully even on tasks it had never seen before by using a multitask learning paradigm. The model was trained on over 40 datasets created from various tasks such as SuperGLUE, WikiQA, and SQuAD, and tested in zero-shot scenarios. In zero-shot experiments on the Natural Questions (NQ) dataset, the model achieved an F1-score of 77.1% and a precision of 74.3%. Despite being language model-focused, the model demonstrated satisfactory performance on tasks requiring token-level answer extraction.

The T5 (Text-to-Text Transfer Transformer) architecture developed by Raffel et al. [33] offers a universal model approach by translating all NLP tasks into a "text-to-text" format. The model was trained multi-task on GLUE and SuperGLUE benchmarks and tested on a wide range of tasks such as NLI, QA, summarization, and translation. In tests conducted on the SQuAD v1.1 dataset, it achieved an F1-score of 89.1% and an EM (exact match) score of 86.5%. T5 provides a strong foundation for token-level inference and semantic matching, and also has zero-shot transfer capabilities across different language pairs.

Liu et al. [34], in their study titled Multi-task Deep Neural Networks for Natural Language Understanding, proposed solving tasks such as NER, sentiment analysis, and paraphrase detection with a single model using a multi-task learning structure built on the same encoder architecture. In their results obtained on the GLUE benchmark, they achieved 89.3% precision, 86.4% recall, and 87.8% F1-score, particularly in the paraphrase detection task. The model's low deviation and high consistency in token-level tasks (e.g., NER) is important for semantic token-level matching tasks.

Yin et al. [35], in their study titled Benchmarking Zero-shot Text Classification, measured zero-shot learning capability by converting text classification tasks into entailment problems. The model was structured with the goal of performing on unlabeled classes from scratch, rather than using directly comparable datasets such as SNLI and MultiNLI. In zero-shot tests conducted on AG News, the model achieved a 70.2% F1-score. Although it performs sentence-level classification, the soft-alignment mechanism used in the model provides a usable basis for token-level interpretation.

Wang et al. [36], in their study titled "Entailment as Few-Shot Learner for Extractive Question Answering" tested token-level answer inference tasks in a few-shot scenario using entailment-based methods. The model, trained on only 16 examples from the SQuAD dataset, achieved a 71.8% F1-score even in zero-shot settings. This approach demonstrates that high performance can be achieved with limited data in semantic alignment tasks, contributing to the reduction of labeling costs.

Reimers & Gurevych [37], in their work titled Making Monolingual Sentence Embeddings Multilingual using Knowledge Distillation, aimed to adapt powerful English-based embedded models (sentence-BERT) to multilinguality. The multilingual BERT model used as the student model was tested on semantic similarity tasks in over 50 languages (including Turkish). Comparisons made on the Tatoeba dataset yielded 84.6% precision, 83.1% recall, and 83.8% F1-score for Turkish. It offers a directly applicable structure for tasks such as token-level semantic alignment specifically for Turkish.

The UnifiedQA model developed by Khashabi et al. [38] aims to solve multi-format QA tasks within a single encoder–decoder framework. The model was trained on token-level extractive QA datasets such as SQuAD, TriviaQA, and Natural Questions, and demonstrated strong performance, particularly in token extraction tasks. It achieved a 78.4% F1-score and 76.0% precision on the Natural Questions dataset. The UnifiedQA architecture, with its multi-task and transfer learning capabilities, can be directly referenced for late-interaction token matching tasks.

Assessments regarding previous studies

Table 1 systematically summarizes the conceptual similarities and differences between the models found in the literature and their boundary conditions, and the proposed "Late-Interaction Fusion" architecture.

Upon examining Table 1, it can be seen that the model we propose is conceptually or structurally similar to these studies in some respects, but also has distinct differences in terms of both task level and architectural components.

In particular, BiDAF [28] shares a similar goal with the proposed Late-Interaction Fusion architecture in that it establishes bidirectional attention between the query and context to enable token-level interaction. However, since BiDAF focuses on extractive question answering tasks, it produces two outputs consisting of "start" and "end" positions for each example and does not use a Transformer-based encoding structure. In contrast, our proposed model can model semantic densities at the token level much more flexibly thanks to its ModernBERT-based Transformer encoder and combines both Bi-Encoder efficiency and Cross-Encoder accuracy in a hybrid manner by processing query–text pairs independently.

The T0 [32] and T5 [33] studies are similar in that they are Transformer-based and aim for semantic generalization, particularly through multitask learning or text-to-text generation structures. However, the task design of these models is mostly in the form of question-answering (QA) or multi-task language generation (multi-choice/generative tasks). Therefore, they focus on sentence or text-level response generation rather than token-level highlighting or query-related word-level matching. In this respect, T0 and T5 are similar to the proposed system in terms of model family, but differ in terms of output format and task level.

The Entailment as Few-Shot Learner (EFL) [36] study aims to achieve generalization with low data by converting each task into an entailment problem. In this respect, it shares conceptual similarities with the proposed system's semi-supervised LLM-based data augmentation approach in terms of generalization with low data. However, while EFL makes "entail/not entail" decisions at the sentence level, our proposed model adopts a token-level multi-label classification structure that determines the relationship of each token within the context to the query using 0/1-based probabilities. Therefore, both approaches draw on the idea of semantic matching, but differ in terms of granularity level and labeling unit.

Similarly, the UnifiedQA [38] study aims to achieve formal generalization by establishing a single QA model across multi-format datasets. Although it shares architectural similarities with the proposed system in terms of its Transformer-based encoder–decoder structure, UnifiedQA's target output is the identification of answer spans within the context.

Table 1
A comparative overview of relevant studies in terms of job descriptions, class boundaries, and structural proximity to the proposed model

Author (Year)	Model	Task	Class	Description
Parikh et al. (2016)	Decomposable Attention	NLI (SNLI)	3 class	Entailment, Contradiction, Neutral
Yin et al. (2016)	ABCNN	Paraphrase Detection (MSRP)	2 class	Paraphrase or Not paraphrase
Wang et al. (2017)	BiMPM	NLI (SNLI) or QQP	3 or 2 class	SNLI → 3; QQP → Duplicate, Non-duplicate
Chen et al. (2017)	ESIM	NLI (SNLI)	3 class	Entailment, Contradiction, Neutral
Seo et al. (2016)	BiDAF	QA (SQuAD)	No class	Span Extraction
Tay et al. (2018)	MCAN	QA matching (TrecQA)	2 class	True or False
Gong et al. (2017)	IoI	NLI (MultiNLI)	3 class	Entailment, Contradiction, Neutral
Guo et al. (2016)	DRMM	Retrieval (TREC Robust04)	2 class	Relevant or Non-relevant
Sanh et al. (2021)	T0	Multitask (SuperGLUE, QA, NQ)	No class	Task-based flexible structure (NLI fixed, QA unlimited)
Raffel et al. (2020)	T5	Multitask (text-to-text)	No class	Unlimited QA & generative tasks
Liu et al. (2019)	MT-DNN	Multitask (NER, sentiment, paraphrase)	Partially	Sentiment/paraphrase fixed, NER flexible
Yin et al. (2019)	Zero-Shot Text Classification	Text Classification (AG News)	4 class	World, Sports, Business, Sci&Tech
Wang et al. (2021)	Entailment Few-Shot QA	Extractive QA (SQuAD)	No class	Span Extraction
Reimers & Gurevych (2020)	Multilingual SBERT	Semantic Similarity (Tatoeba)	2 class	Similar or Not similar
Khashabi et al. (2020)	UnifiedQA	Multi-format QA (SQuAD, NQ, TriviaQA)	No class	Extractive QA

Therefore, the system is focused on answer generation. The proposed model, on the other hand, is geared toward the semantic token-level matching task of identifying words or sub-units related to the query in Turkish text.

In conclusion, some of the studies in the literature (BiDAF, UnifiedQA) share similarities with the proposed model in terms of interaction strategy, some (T0, T5) in terms of Transformer architecture, and some (EFL) in terms of the idea of semantic generalization. However, none of them utilize a ModernBERT-based Late-Interaction Fusion architecture to model the relationships between queries and text at the token level in a morphologically complex language such as Turkish, nor do they possess a holistic structure trained with LLM-supported semi-supervised data augmentation. In this respect, our work differs from approaches in the literature in terms of both task definition and architectural design, while also drawing conceptually from these previous studies.

In light of these evaluations, the proposed system presents a comprehensive structure by leveraging the strengths of various approaches in the literature. Thus, our work synthesizes the partial contributions of previous studies into a single structure, proposing a new paradigm that simultaneously provides both token-level meaning matching and a high efficiency–accuracy balance. In this respect, the study provides a unique foundation for the development of advanced semantic highlighting systems for Turkish.

Material and methodology

This section provides details about the dataset used in the study and the preparations leading up to model training. Fig. 1 illustrates the construction of the base dataset and the subsequent stages of the pipeline, from data collection and the selection of the annotator model through LLM-assisted labeling and tokenisation to the training of the proposed architecture.

Collecting dataset

The dataset created in this study consists of approximately 500,000 different texts compiled from various sources such as news articles, technical blog posts, product reviews, and forum posts. This main data was labeled using an LLM-supported semi-supervised approach. As a result of this labeling process, approximately 2 million labeled query–text pairs were obtained. The aim is to prevent the dataset from developing narrow expertise focused solely on a specific industry or product type. To this end, customer support requests, user comments, and complaint texts related to products or services have been compiled from diverse fields such as finance, e-commerce, telecommunications, and the service sector.

The main reason for choosing these types of data is that users express a problem or requirement in natural language in the form of an "implicit query" and the rest of the text contains evidence or explanatory details related to this query. This feature provides a natural testing environment for the "semantic token matching" task. Such a focus has ensured that the developed model is no longer a purely theoretical approach, but rather a practical solution with high industrial value that can handle the complexity of real-world data.

Selecting LLM for labeling

LLMs are proving to be a highly effective tool in data augmentation processes, particularly in situations where obtaining labeled data is costly. In the literature, studies on the use of LLMs for this purpose generally fall into two main trends [39,40]. The first trend focuses on increasing data diversity by generating entirely new and synthetic text examples. The second trend positions LLMs as data annotators or reasoners on existing real-world data and aims to leverage these models' automatic labeling/meaning extraction capabilities. The primary goal of adopting the second trend in this study is to position the LLM not as a "content generator" but as a large-scale, high-accuracy zero-shot data annotator. The success of LLMs in such tasks has been demonstrated in pioneering studies [41]. Before proceeding with the labeling process, approximately 1% of the data, representing the entire dataset in the best possible way, was manually labeled by expert human labelers. Different LLM models were tested on this small dataset, referred to as the gold standard. Certain types of LLMs (e.g., GPT-4.1 and GPT-4o) were not preferred in this study because they cannot be run on local servers and, if used on a paid basis, would increase the economic cost of the work. Among the LLM models that could overcome these limitations, the model named Qwen3-32b yielded the most successful results. After selecting the LLM, the process of LLM-assisted labeling was initiated.

The protocol behind this selection is as follows. The gold standard corresponds to approximately one percent of the base corpus, sampled to preserve the distribution of the source domains, and was annotated by four expert annotators working from a common written guideline; the set was partitioned among them, so each was responsible for a disjoint portion and worked independently. Each candidate LLM was run on this set with the same prompts later used for the full corpus, and the reported figure is the proportion of individual annotation decisions on which model and human reference coincide, computed at the level of decisions rather than documents. Two limitations follow. No inter-annotator agreement coefficient can be reported, since the disjoint

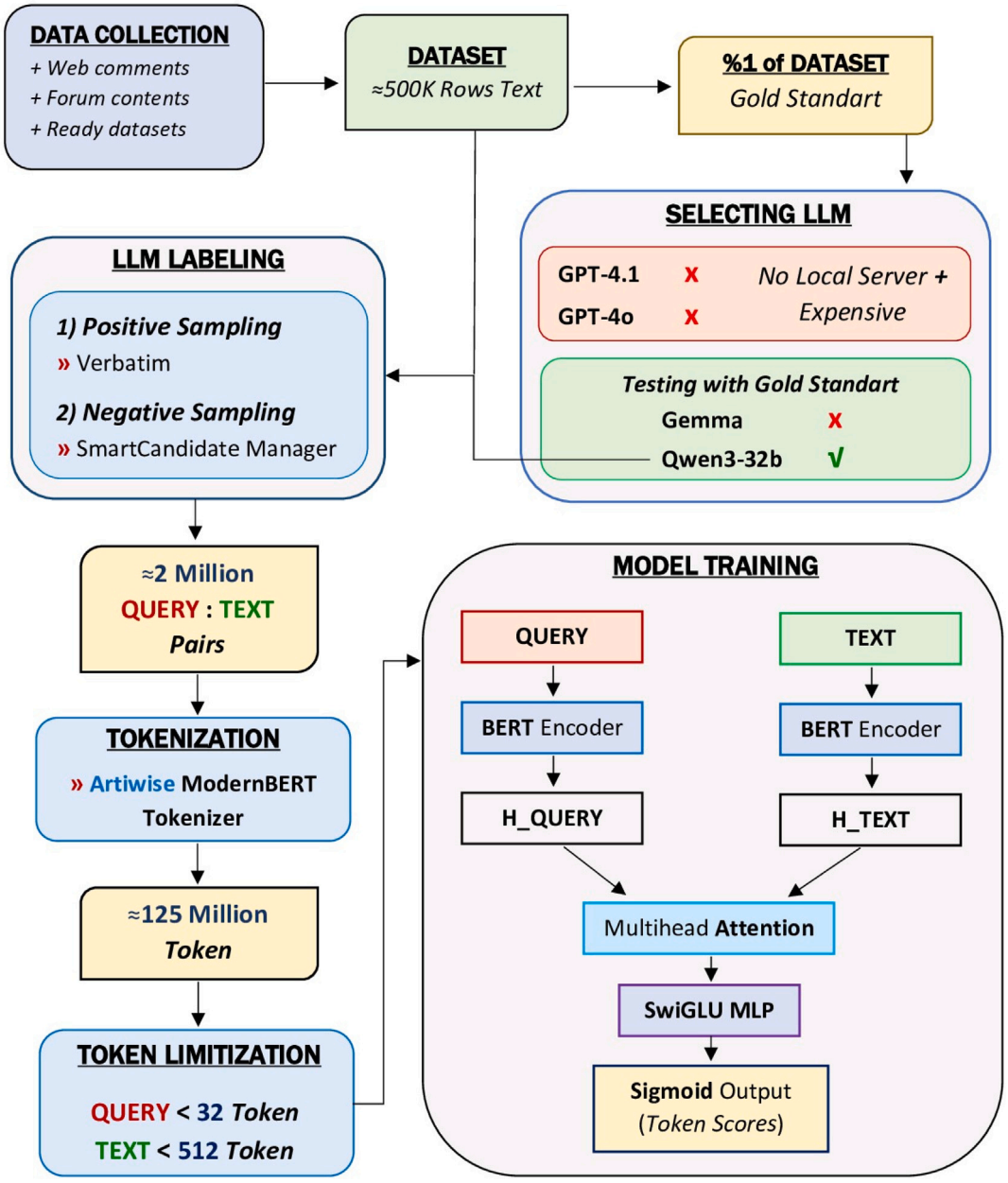


Fig. 1. Overall pipeline of the study: collection of the base corpus, manual annotation of the gold standard subset, selection of the annotator LLM, two-stage LLM-assisted labeling, tokenisation and length limitation, and training of the proposed late-interaction architecture.

partition means no text was annotated more than once and a measure such as Fleiss' κ requires overlapping annotations by construction. The 92% figure also measures agreement rather than absolute correctness, and therefore bounds the label quality attainable by the automatically annotated portion of the corpus.

LLM-based data labeling

A novel two-stage sampling strategy (positive and negative sampling) was used for the LLM-supported data labeling process. To ensure the reproducibility and transparency of the process, some basic prompt structures were also employed. In the positive sampling phase, the aim is for the LLM to identify all the different topics within a given text on its own, using a zero-shot approach, without the need for any predefined category list [35]. The most critical aspect of this method is the requirement that the model must provide evidence showing where each identified topic appears in the text by verbatim quoting from the text [42]. Thus, the developed evidence-based annotation approach ensures that the generated topics are directly grounded in the text and also safeguards data integrity. This technique also increases the transparency and verifiability of the annotation process [43]. Table 2 shows the prompt used for positive sampling on the selected LLM.

When examining the prompt structure presented in Table 2, it is evident that the task undertaken by the LLM during the positive sampling phase is not limited to topic extraction alone, but also includes the function of directly identifying and reporting the contextual evidence for each topic within the text. The first rule of the prompt requires the model to extract all possible different topics, while the second rule emphasizes the requirement to "verbatim" quote the in-text reference for each extracted topic. This encourages the LLM to demonstrate strong awareness not only at the semantic level but also at the surface level of the text. The formal structure of the example prompt in the table is also noteworthy. The requirement for JSON format output increases both repeatability and programmatic analysis capabilities by

Table 2
Prompt used in LLM-supported positive sampling processes.

Task	Prompt

Positive Sampling	Analyze the following customer review and generate a response in JSON format according to the specified rules.
	RULES
	1. Topic Extraction: Identify all distinct topics mentioned in the review.
	2. Segment Addition: For each topic, add the sentences or phrases where that topic is mentioned to the "segments" list by copying them verbatim from the review text.
	Important: Segments must never be summaries or interpretations.
	Required JSON format
	{
	''categories'':[
	{''category'':''<topic_name>'',
	''segments'':[''<segment_1>'',''<segment_2>'']
	}
	]
	}
	Customer Review
	{content}

Table 3
Prompt used in LLM-supported negative sampling processes.

Task	Prompt

Negative Sampling	From the list of categories below, determine which ones are actually present in the customer review.
	Candidate List
	{ ', '.join(negative_candidates) }
	Return **only** the following JSON format.
	{
	''present'':[''<present_category1>'',
	''<present_category2>'']
	}
	Customer Review
	{content}

converting the model's responses into a standard data structure. This facilitates the direct integration of LLM outputs into data pipelines or label management systems. In addition, the fact that the rule set is given in the form of "strict enforcement" (e.g., "Failure to follow this rule will render the output completely invalid".) is a measure that strengthens the model's behavioral stability.

In the negative sampling phase, based on the fact that the model must also learn what a text "is not" an indirect method was adopted to generate high-quality negative labels [44]. In this process, candidate categories with a low probability of being related to the text were selected by the SmartCandidateManager module integrated into the system and presented to the LLM. The model was asked to indicate which of these candidate categories were "present" in the text. Subsequently, according to system logic, categories not marked as present by the LLM were accepted as verified negative labels. This indirect strategy is based on the assumption that confirming the presence of a concept is a more cognitively reliable task for the LLM than confirming its absence [45]. Table 3 shows the prompt used for negative sampling on the selected LLM.

The prompt structure in Table 3 clearly reflects the conceptual goals and operational logic of the negative sampling process. This structure ensures that the model indirectly derives negative examples through the principle of positive confirmation, rather than directly predicting "negative" categories. In other words, the LLM is provided with a list of potential candidate categories and is only asked to mark those that are actually "present" in the text. Thus, the categories not selected by the model are automatically considered confirmed negative labels.

When examining the formal design of the prompt, it is evident that the guidance is created in a highly minimal and error-tight manner. The phrase "Return ONLY the following JSON" is intended to prevent the model from generating explanations or justifications, directing the output entirely toward a structured format. This approach reduces noise during the labeling process and enables LLM outputs to be directly integrated into negative example pools. Furthermore, the fact that the JSON format is organized around a single field, present, facilitates the automatic reversal of the output. That is, categories not included in the present list can be directly associated with the "absent" or "negative" label.

Another important aspect of the prompt in Table 3 is the involvement of the SmartCandidateManager module during the pre-selection phase. This module filters candidate categories to be presented to the LLM in a way that minimizes their likelihood of being related to the text. This reduces the cognitive load on the model while making the selection of "truly existing" categories more distinct. This structure represents an early form of the LLM-human hybrid labeling paradigm in the indirect negative sampling process. This is because the system produces reliable negative data by interpreting the model's positive decisions in reverse, rather than directly predicting possible negatives that would otherwise require human expert determination.

Artiwise ModernBERT language model

The selection of the language model used in this study has been evaluated as a critical design decision that directly impacts the overall success of the project. Although numerous BERT-based models [11] have been developed for Turkish, "Artiwise ModernBERT" [46] was chosen as the base model for this study. This model is based on the ModernBERT architecture introduced by Warner et al. [19], which significantly improves the classic BERT architecture with contemporary optimization techniques.

The benchmark results of the model developed by ArtiwiseAI, available on Hugging Face, demonstrate its success in Turkish. In various Turkish natural language understanding (NLU) tasks, it has shown consistent, statistically significant, and markedly higher performance gains compared to other widely used Turkish BERT variants [47,48]. This demonstrates that the fundamental architectural improvements offered by ModernBERT have been successfully transferred to the Turkish language.

ModernBERT offers significant improvements in both speed and memory usage compared to the standard BERT architecture. This makes training (fine-tuning) and inference processes more efficient, especially on large datasets. While traditional BERT models are limited to a context window of 512 tokens, the ModernBERT architecture naturally increases this limit to 8192 tokens. Although the standard 512 token limit was used in this study, the model's natural long-context capacity offers fundamental potential for capturing semantic relationships in more complex and longer texts.

In this study, the pretrained Artiwise ModernBERT model was not used directly. The model was fine-tuned using the "Late-Interaction Fusion" architecture on an enriched dataset developed specifically for this study, containing approximately 2 million query–text pairs. This process adapted the general-purpose model, also presented in the paper by Warner et al. [19], for this niche task by focusing it on the specific task of token-level semantic matching.

Tokenization

All queries and texts were split into subwords using the WordPiece tokenizer of the Artiwise ModernBERT language model, which served as the basis for this study. This approach effectively managed the word diversity arising from the agglutinative nature of the Turkish language. The final dataset created contains approximately 2 million query–text pairs obtained through the LLM-supported labeling process of the texts collected from the aforementioned sources. The token-level labeling operations performed on these pairs produced a total of approximately

Table 4
Principal statistics of the dataset.

Property	Value
Base texts collected	500,000
Query-text pairs after annotation	≈2,000,000
Labeled tokens	≈125,000,000
Distinct query expressions	$\approx 5 \times 10^5$
Average queries per text	≈4
Average text length	≈62 tokens
Average query length	≈3.1 words
Maximum text length	512 tokens
Maximum query length	32 tokens
Positive token ratio	≈7.5%
Gold standard (manually annotated)	≈1% of base texts
Expert annotators	4 (disjoint portions)
Encoder vocabulary size	50,358

125 million labeled tokens. Additionally, the length of each text segment has been limited to a maximum of 512 tokens in order to balance the model's processing capacity and maintain contextual consistency. Queries have also been limited to a maximum of 32 tokens.

The principal statistics of the resulting dataset are summarized in Table 4. Two figures in that table deserve comment. The average text length of approximately 62 tokens follows from the ratio between the 125 million labeled tokens and the two million query–text pairs, and shows that the 512-token limit is generous relative to the typical input rather than a binding constraint. The average of about four queries per text reflects the two-stage annotation strategy, in which each text receives both the topics that the annotator model identified in it and a set of indirectly verified negative categories.

The query space of this dataset differs from that of a conventional classification corpus. Because the annotator model identified the topics of a text in a zero-shot manner, without a predefined category list, the query set is open-ended rather than drawn from a fixed inventory: queries are short, free-form noun phrases averaging 3.1 words, such as "customer service", "product quality" and "non-refund of the deposit", and the number of distinct expressions is on the order of 5×10^5. This figure is approximate and is inflated by near-duplicate expressions that differ only in capitalization or wording, as no canonicalization step was applied. Their frequency distribution is strongly long-tailed, a small number of generic categories accounting for a disproportionate share of the annotations while most expressions occur only once or twice. This open vocabulary is what makes the task a zero-shot matching problem rather than a closed-set classification problem.

At the token level the label distribution is heavily skewed towards the negative class, since only the segments quoted as evidence for a query are marked positive and every indirectly verified negative category contributes a text in which no token is positive at all. Positive tokens account for approximately 7.5% of all labeled tokens, a negative-to-positive ratio of roughly twelve to one, which is the direct motivation for the Focal Loss component of the training objective described in Section "Model Architecture and Training".

Model architecture and training

Zero-Shot Token Classification (ZSTC) research [35] presents a number of challenges, including large-scale data labeling, rapid model testing, and managing the complex interaction between LLM and the model. To overcome these challenges, the study utilized an integrated web-based platform called ZSCAT (Zero-Shot Classification Annotation Tool). All processes within the study, including data preparation, annotation, model training, and analysis, were carried out on this platform. The proposed model is based on a hybrid approach that aims to strike a balance between efficiency-focused Bi-Encoder architectures and performance-focused Cross-Encoder architectures in the literature. The general architecture of the model is shown in Fig. 2.

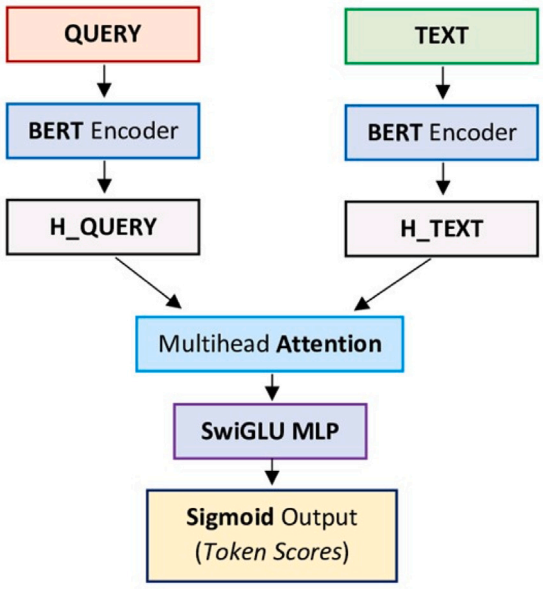


Fig. 2. Architecture of the model proposed in the study.

The model architecture consists of three stages: Bi-Encoder, Late-Interaction Fusion, and Token Classification. In the Bi-Encoder stage, the query and text are processed separately, completely independently of the ModernBERT encoder. This technique allows each to have its own contextual embeddings calculated. This approach allows text representations to be pre-computed and stored. This results in extremely high inference efficiency, especially when searching large document sets.

In the Late-Interaction Fusion stage, the query and text representations (H_QUERY and H_TEXT) obtained in the first stage are fed into a special "late-interactive" fusion layer. This layer uses a standard Transformer cross-attention mechanism (MultiheadAttention) to enable text tokens to "attend" to query tokens. This creates a rich, context-sensitive representation (H_FUSED) of which text tokens are semantically most relevant to the query. The formulaic representation of H_FUSED is as shown in Eq. (1).

$$\begin{aligned}\tilde{H} = \text{MultiheadAttention}(&\text{query} = H_{\text{TEXT}},\\ &\text{key} = H_{\text{QUERY}},\\ &\text{value} = H_{\text{QUERY}})\end{aligned} \tag{1}$$

Detailed formulation of the fusion stage

Let $H_{\text{QUERY}} \in \mathbb{R}^{L_q \times d}$ and $H_{\text{TEXT}} \in \mathbb{R}^{L_t \times d}$ denote the contextual representations produced independently by the shared ModernBERT encoder, where $d = 768$, $L_q \leq 32$ and $L_t \leq 512$. The asymmetric assignment of roles in Eq. (1) is deliberate: the text supplies the *queries* of the attention, so the output preserves the length and ordering of the text, which is a prerequisite for producing one decision per token.

For each attention head $h \in \{1, \ldots, H\}$ with $H = 8$ and head dimension $d_h = d/H = 96$, the projections are computed as given in Eq. (2).

$$\begin{aligned}Q^h &= H_{\text{TEXT}} W_Q^h,\\ K^h &= H_{\text{QUERY}} W_K^h,\\ V^h &= H_{\text{QUERY}} W_V^h,\end{aligned} \tag{2}$$

where $W_Q^h, W_K^h, W_V^h \in \mathbb{R}^{d \times d_h}$ are learned parameters that are not shared with the encoder. The attention distribution of each text token over the query tokens is then obtained as in Eq. (3).

$$A^h = \text{softmax}\left(\frac{Q^h (K^h)^\top}{\sqrt{d_h}} + M\right) \tag{3}$$

Here $M \in \mathbb{R}^{L_t \times L_q}$ is an additive mask that sets the columns of padded query positions to $-\infty$, so that a short query is treated identically regardless of batch composition. The head outputs are concatenated and projected back to the model dimension as in Eq. (4).

$$\tilde{H} = \mathrm{Concat}\left(A^1 V^1, \ldots, A^H V^H\right) W_O \tag{4}$$

Dropout with rate 0.1 is applied to the output, giving Eq. (5). The block contains no residual connection around the attention output and no layer normalization before or after it; adding a pre-norm residual path is listed among the ablations proposed in Section "Conclusion".

$$H_{\mathrm{FUSED}} = \mathrm{Dropout}(\tilde{H}) \tag{5}$$

Each row of H_{FUSED} is a combination of query-side value vectors weighted by the alignment of that particular text token, so two tokens aligning differently with the query receive different representations.

In the Token Classification stage, the enriched text representations H_{FUSED} are passed through a position-wise feed-forward block with "SwiGLU" activation [49], whose gated form is given in Eq. (6).

$$\mathrm{FFN}(x) = \left(\mathrm{SiLU}(xW_1) \odot xW_2\right) W_3 \tag{6}$$

In Eq. (6), $W_1, W_2 \in \mathbb{R}^{d \times d_{ff}}$ and $W_3 \in \mathbb{R}^{d_{ff} \times d}$ with an inner dimension of $d_{ff} = 2048$, and $\odot$ denotes element-wise multiplication. The gate lets the layer suppress dimensions irrelevant to the current query rather than applying a fixed non-linearity to all of them. A linear scoring head followed by a sigmoid then produces the final per-token probability, as expressed in Eq. (7).

$$\begin{aligned} H_{\mathrm{OUT}} &= \mathrm{FFN}\left(H_{\mathrm{FUSED}}\right), \\ P_i &= \sigma\left(w^\top H_{\mathrm{OUT},i} + b\right) \end{aligned} \tag{7}$$

This process generates a score in the range $[0, 1]$ for each token i of the text, representing the likelihood of relevance to the query, and transforms the problem into a token-based binary classification task. A token is reported as relevant when $P_i \geq 0.5$; this operating point was kept fixed for all models so that the comparison in Section "Results and Discussion" is not influenced by threshold tuning.

The fusion stage adds approximately 7.1M parameters, about 5% of the encoder, and its cost is linear in both sequence lengths because the attention matrix is $L_t \times L_q$ rather than $(L_t + L_q)^2$ as in a Cross-Encoder. For $L_t = 512$ and $L_q = 32$ an analytical operation count gives roughly 3.1 GMAC per fusion pass against 43 GMAC for a full 12-layer encoder pass, so evaluating an already-encoded text against a further query costs less than a tenth of the encoding itself.

Because each row of A^h is a normalized distribution of a text token over the query tokens, it can be inspected to explain why a token was marked relevant; this is the basis of the interpretability claim, and a Bi-Encoder comparing two pooled vectors cannot provide it.

The approach adopted in this study was selected based on a strategic tradeoff analysis, considering the advantages and disadvantages of the Bi-Encoder [10] and Cross-Encoder [50] architectures. A pure Cross-Encoder architecture promises the highest performance because it combines the query and text from the outset, establishing deep interaction at every Transformer layer. However, this approach is too slow for practical applications and large text corpora, as it requires running the entire model from scratch for each query–text pair. A pure Bi-Encoder, at the opposite extreme, encodes the query and the text independently and therefore allows text representations to be computed once and reused indefinitely. The downside of this efficiency is that all interaction between query and text is established at the very end, typically through a simple cosine similarity, and thus misses semantic nuances.

The consequences of this tradeoff can be quantified with the operation counts established in Section "Detailed Formulation of the Fusion Stage". Consider a scenario in which 10 different queries are to be evaluated against 1000 different texts of 512 tokens each. A Cross-Encoder must run the entire Transformer from scratch for every pair, that is $10 \times 1000 = 10{,}000$ full encoder passes, corresponding to roughly 430 TMAC. The proposed model instead encodes the 1000 texts and the 10 queries once (1010 encoder passes, of which the query passes are almost free because a query is at most 32 tokens) and then executes only the lightweight fusion head for each of the 10,000 pairs, giving approximately $43 + 31 \approx 74$ TMAC in total, i.e. close to a sixfold reduction. The gap widens further in a deployed system: since text representations are computed offline and stored in a vector database, the encoding term disappears from the online path altogether and the per-query cost reduces to the 31 TMAC of the fusion head, roughly a fourteenfold saving with respect to the Cross-Encoder. It should be noted that, unlike a pure Bi-Encoder, the proposed architecture does not reduce the online cost to a single similarity computation, because the fusion layer must be executed once per query–text pair; the essential point is that this per-pair cost is an order of magnitude smaller than a full encoder pass, which is what makes the design practical at scale.

The hybrid model proposed in this study aims to combine the best aspects of both worlds. While preserving efficiency potential by processing representations separately, like Bi-Encoder, the rich interaction capability of CrossEncoders is replicated through a lightweight yet powerful 'MultiheadAttention' fusion layer added at the end. This allows the model to achieve a balance point that is both powerful in terms of performance and scalable for practical applications.

Our architecture follows a late-interaction paradigm inspired by models such as ColBERT, enabling efficient token-level matching while preserving interaction capacity [13]. The late-interaction approach aims to mimic the rich interaction capabilities of Cross-Encoders through a lightweight yet effective merging mechanism applied in the final stage, while preserving the computational efficiency (pre-computable representations) of Bi-Encoder architectures. This technique enables searching within milliseconds, even in industrial-scale systems containing billions of documents.

One of the most fundamental challenges encountered during the token classification stage is the extreme imbalance between tokens labeled as positive (relevant) and negative (irrelevant). The number of negative examples far exceeds that of positive ones. This situation carries the risk of dominating the model's training towards the overwhelming majority of "easy negatives" when using a standard loss function, preventing the model from learning the challenging but important positive examples.

To solve this problem, a training strategy has been developed based on the Focal Loss [21] function, a widely used method for training under severe class imbalance. Focal Loss enhances the standard Binary Cross-Entropy (BCE) loss by adding a modulation factor that dynamically scales it according to the model's confidence in its prediction. This function is an enhanced version of Cross-Entropy (CE) Loss and balances the model's learning by reducing the impact of easy examples and giving more weight to difficult examples. This allows training to automatically focus on the "difficult" examples. In this study, a hybrid combination of BCE and Focal Loss was used. The formulaic representation of this combination is given in Eq. (8).

$$L_{\mathrm{TOTAL}} = L_{\mathrm{BCE}} + \lambda\, L_{\mathrm{FOCAL}} \tag{8}$$

The λ value in Eq. (8) is a hyperparameter that balances the effect of Focal Loss on the total loss.

Learned parameters and hyperparameters

Model training was conducted using distributed training on two NVIDIA H100 GPUs. This infrastructure enabled the model to be trained on a large dataset containing approximately 2 million examples within a reasonable timeframe. Distributed training was implemented using PyTorch DistributedDataParallel (DDP) [51], and memory optimization was achieved using the gradient accumulation technique. Thanks to this set-up, the fine-tuning process of the ModernBERT-base model, which has approximately 136M parameters, was completed efficiently. The

Table 5
Hyperparameters used in model training.

Hyperparameter	Value
Learning Rate	2e−5
Optimizer	AdamW [52]
Warmup Steps	1000
Weight Decay	0.01
Batch Size	32 (16 per GPU)
Epochs	10 (Best checkpoint: 2)
Max Query Length	32
Max Text Length	512

Table 6
Artiwise ModernBERT model parameters.

Parameter	Value
Model Size	136 M
Hidden Dim	768
Attention Heads	12
Encoder Layers	12
Vocabulary Size	50 358

key hyperparameters used during the training process are summarized in Table 5.

The model parameters of the Artiwise ModernBERT model, which is the base model in the study, are also summarized in Table 6.

The parameters of the late-interaction fusion layer in the proposed model are given in Table 7.

Reference models

The study contains innovations in many respects. As detailed in the section titled "Assessments Regarding Previous Studies" although studies in the literature resemble this study in some respects, they contain differences that cannot be fully compared. Therefore, in order to fairly measure the success of the model proposed in this study and to compare the results obtained with the fundamental approaches in the literature, reference models of varying levels of complexity were defined. Subsequently, the results obtained by these models using the same dataset and the results obtained by the model proposed in this study were evaluated in the "Results and Discussion" section.

Keyword-phrase matching

This method is an advanced version of traditional keyword search methods. Although it lacks semantic understanding capabilities, it provides a strong lexical reference point. To use this method, both the input text and the search query are first stripped of punctuation marks and converted to lowercase, then split into words. Then, each word is reduced to its morphological root (Stemming) using the Zemberek library. The Zemberek library has been used in many scientific studies from the past to the present, especially in "Turkish NLP" tasks [53,54]. Finally, the stemmed word sequence of the query is searched for as a whole (in sequential order) within the stemmed word sequence of the text. Algorithm 1 contains pseudo-code for Keyword-Phrase Matching.

Algorithm 1 : Keyword-Phrase Matching

```
Algorithm 1 Keyword-Phrase Matching.
FUNCTION KEYWORD_PHRASE_MATCH(Text, Query)
BEGIN FUNCTION
    SET T_norm TO (CALL NORMALIZE with Text)
    SET Q_norm TO (CALL NORMALIZE with Query)

    SET TOK_T TO (CALL TOKENIZE with T_norm)
    SET TOK_Q TO (CALL TOKENIZE with Q_norm)
    SET STEM_T TO (CALL STEM with TOK_T)
    SET STEM_Q TO (CALL STEM with TOK_Q)

    SET n TO (CALL LENGTH with STEM_T)
    SET m TO (CALL LENGTH with STEM_Q)
    SET L TO (CALL ZERO_ARRAY with LENGTH(TOK_T))

    IF m EQUALS 0 THEN
        RETURN L
    ENDIF
    IF m > n THEN
        RETURN L
    ENDIF

    FOR i FROM 0 TO (n - m) DO
        SET match_flag TO TRUE
        FOR k FROM 0 TO (m - 1) DO
            IF STEM_T[i + k] NOT EQUALS STEM_Q[k] THEN
                SET match_flag TO FALSE
                BREAK
            ENDIF
        ENDFOR

        IF match_flag EQUALS TRUE THEN
            FOR j FROM 0 TO (m - 1) DO
                SET L[i + j] TO 1
            ENDFOR
        ENDIF
    ENDFOR
    RETURN L
END FUNCTION
```

Dense retrieval

This reference model, which forms the basis of modern NLP systems, goes beyond lexical matching of words to measure semantic proximity. In this method, the Artiwise ModernBERT model was used to generate semantically rich vector representations (dense embeddings) for each token in both the query and the text. As a comparison mechanism, the cosine similarity between the vector of the token representing the entire query and the vector of each token in the text is calculated. In order to detect relationships above a certain semantic proximity, in line with common assumptions in the literature, tokens with a similarity score above the threshold value of 0.7 were considered a "match". This method provided a strong reference point for comparing the performance of the proposed model with a standard semantic search technique. Algorithm 2 contains pseudo-code for Dense Retrieval.

Algorithm 2 : Dense Retrieval

```
Algorithm 2 Dense Retrieval.
FUNCTION DENSE_RETRIEVAL(Text, Query, Encoder, Theta)
BEGIN FUNCTION

    SET T_enc TO (CALL Encoder.Encode with Text)
    SET Q_enc TO (CALL Encoder.Encode with Query)

    SET V_query TO (CALL Q_enc.get_cls_embedding)
    SET V_text_tokens TO (CALL T_enc.get_token_embeddings)
    SET N TO (CALL LENGTH with V_text_tokens)

    SET L TO (CALL ZERO_ARRAY with N)

    FOR i FROM 0 TO (N - 1) DO
        SET V_token TO V_text_tokens[i]
        SET sim TO (CALL COSINE_SIM with V_query, V_token)

        #   Theta: similarity threshold (e.g., 0.7)
        IF sim > Theta THEN
            SET L[i] TO 1
        ENDIF
    ENDFOR

    RETURN L
END FUNCTION
```

Standard Bi-Encoder

This reference model independently encodes the query and text using the same language model (Artiwise ModernBERT), producing semantic representations (embeddings) for each. The resulting query representation (H_QUERY) and text representation (H_TEXT) become

Table 7
Parameters of the late-interaction fusion layer in the proposed model.

Parameter	Value
Attention Heads	8
Hidden Dim	768
Dropout Rate	0.1
MLP Hidden Size	2048
Activation Function	SwiGLU

comparable in the same vector space. The semantic similarity between these representation pairs is calculated using the cosine similarity metric. This determines the extent to which the query overlaps with the text. This method works quickly on large datasets due to its high inference efficiency. However, since the Standard Bi-Encoder does not establish direct interaction between the query and the text, it is limited in capturing context-sensitive fine-grained semantic relationships. In this respect, it differs from the proposed baseline model. The proposed model aims to capture contextual meaning more deeply by adding a "Multihead Attention"-based late-interaction layer between the two representations in the final stage to address this weakness of the Bi-Encoder architecture. Pseudo-code for the Standard Bi-Encoder is provided in Algorithm 3.

Algorithm 3 : Standard Bi-Encoder.

```
Algorithm 3 Standard Bi-Encoder.
FUNCTION STANDARD_BI_ENCODER(Text, Query, Encoder)
BEGIN FUNCTION
    SET T_enc TO (CALL Encoder.Encode with Text)
    SET Q_enc TO (CALL Encoder.Encode with Query)

    SET H_TEXT  TO (CALL T_enc.get_pooled_embedding)
    SET H_QUERY TO (CALL Q_enc.get_pooled_embedding)

    SET S TO (CALL COSINE_SIM with H_QUERY, H_TEXT)
    # scalar in [-1, 1] or [0, 1]
    RETURN S
END FUNCTION
```

GLiNER

The fourth reference model is GLiNER [55]. The three baselines above are deliberately positioned as controlled lower bounds on the lexical-to-semantic axis rather than as contemporary competitors; GLiNER was added as the single strong, recent upper reference, for four reasons. First, its output granularity coincides with the task addressed here, so its predictions can be projected onto token labels and evaluated with the same metric and threshold, without an arbitrary post-processing step. Second, it is conditioned on free-form label descriptions rather than a fixed category inventory, which mirrors the open-vocabulary queries used here. Third, it occupies the opposite end of the accuracy-efficiency spectrum from the Bi-Encoder baseline, and is therefore the appropriate yardstick for the central claim that a late-interaction design can approach such accuracy at a fraction of the cost; a comparison against an unconstrained Cross-Encoder would have confirmed the accuracy gap while leaving the efficiency claim untested. Fourth, it is open source and could be retrained under identical data, seed and hardware conditions.

GLiNER was not used as an off-the-shelf pretrained model. It was fine-tuned on the same training corpus, with the same partition, random seed and hardware as all other systems, and evaluated with the same token-level metric and threshold. One deviation from its default configuration was required to keep training feasible and is stated explicitly: the maximum span length was capped at 12 tokens and the span enumeration loop advanced with a stride of 2. This restriction may understate GLiNER's attainable performance slightly, but follows directly from the combinatorial cost that motivates the comparison.

Beyond the motivation for its inclusion, GLiNER also exhibits significant design differences from the proposed model in terms of performance, efficiency, and computational cost, and these differences directly affect the practical applicability of the models. The architecture of GLiNER inherently has a high computational cost. The fundamental reason for this is that the model's operating principle is based on combinatorial span generation [56]: for a text of N tokens and a maximum span length k, up to $N \times k$ candidate spans must be scored, so the cost grows with the number of enumerated spans rather than with the sequence length alone. The computational cost of the proposed model, in contrast, depends only on the length of the input sequence and is dominated by dense matrix multiplications that are highly optimized on GPUs, with no combinatorial term. This design choice ensures that the proposed model is significantly faster and requires fewer resources than GLiNER in both training and inference processes. Pseudo-code for GLiNER is given in Algorithm 4, and Table 8 compares the model proposed in this study with all reference models in several aspects.

Algorithm 4 : GLiNER.

```
Algorithm 4 GLiNER.
FUNCTION GLINER_INFERENCE(Text, LABEL_DESC, Encoder, max_span_len,
     theta_span)
BEGIN FUNCTION
    SET TOK_T TO (CALL TOKENIZE with (CALL NORMALIZE with Text))
    SET N TO (CALL LENGTH with TOK_T)
    SET H_TEXT_TOK TO (CALL Encoder.EncodeTokens with TOK_T)
    SET H_LABEL TO (CALL Encoder.Encode with
         LABEL_DESC).get_pooled_embedding
    SET L TO (CALL ZERO_ARRAY with N)

    FOR start FROM 0 TO (N - 1) DO
        FOR span_len FROM 1 TO (CALL MIN with max_span_len, N -
             start) DO
            SET end TO start + span_len - 1
            SET h_span TO (CALL BUILD_SPAN_REPR with H_TEXT_TOK,
                 start, end)

            SET Z TO (CALL CONCAT with h_span, H_LABEL)
            SET S TO (CALL MLP_SCORE with Z)
            SET P TO (CALL SIGMOID with S)

            IF P >= theta_span THEN
                FOR t FROM start TO end DO
                    SET L[t] TO 1
                ENDFOR
            ENDIF
        ENDFOR
    ENDFOR
    RETURN L
END FUNCTION
```

Table 8 shows that the model proposed in this study offers a solution that balances the efficiency of Bi-Encoder with Cross-Encoder-like semantic interaction. However, the same table also clearly shows that GLiNER is contextually powerful but computationally expensive.

Results and discussion

The proposed model and the previously mentioned reference models were run using the same dataset and yielded results consistent with the qualitative characterization presented in Table 8. The proposed model clearly outperformed the lexical and vector-similarity reference models, and achieved a substantial improvement over the Standard Bi-Encoder architecture, while remaining slightly below the computationally far more expensive GLiNER in terms of F1-score. The results of the experiments conducted in this study are presented in Table 9.

The evaluation protocol is identical for all systems. Predictions and reference labels for every valid, non-padding token are pooled into a single global list, and precision, recall and F1-score are computed over that list for the positive class, at a fixed threshold of 0.5 that was not tuned per system. For each model the checkpoint is the one at the epoch with the highest F1-score on the held-out split. Since the same split is used both to select the checkpoint and to report the figures, the absolute

Table 8
Comparison of the proposed model and reference models in the study.

Feature	Keyword phrase matching	Dense retrieval	Standard Bi-Encoder	GLiNER	Proposed model
Semantic Comprehension Ability	None (Lexical)	Medium (Vectoral)	Medium (Vectoral)	High	High (Context-sensitive)
Contextual Interaction Level	None	Low	Low	Very High	Medium–High (Late-interaction)
Efficiency, Extraction Rate	Very High	High	Very High	Low	High
Computational Cost	Very Low	Low	Low	Very High	Medium (Optimize GPU operations)
Pre-Calculation	Not necessary	Partially	Possible	Not Possible	Possible
Token-Based Output Generation	No	No	No	Yes	Yes
Model Complexity	Very Simple	Simple	Medium	High	Medium–High
Area of Suitability	Simple word searches	Semantic search	Quick semantic matching	Full-scope NER tasks	Real-time semantic matching
Advantage	High speed, low cost	Semantic representation	Scalability	Contextual accuracy	Balanced (Successful + Efficient)
Disadvantage	Semantic blindness	Shallow context	Non-interactive structure	High cost	Medium complexity

Table 9
Comparative table of model metrics.

Model	Precision	Recall	F1-Score
Keyword-Phrase Matching	0.578	0.034	0.065
Dense Retrieval	0.070	0.846	0.129
Standard Bi-Encoder	0.580	0.410	0.481
GLiNER	0.600	**0.560**	**0.580**
Proposed Model	***0.635***	0.496	0.557

values in Table 9 carry the optimistic bias inherent in that design; the relative ordering is unaffected, as the identical procedure was applied to every system.

As shown in Table 9, model architectures with different architectural approaches were evaluated on the same dataset and compared across multiple metrics. The results clearly demonstrate how each model's structural design directly impacts measurement performance. Keyword-Phrase Matching lacks semantic generalization capability because it relies solely on superficial word matching. This results in high Precision (0.578) but very low Recall (0.034). Since the model only flags examples where query words are "exactly matched" as positive, it misses most possible variations, causing the F1-Score to remain at 0.065. This result once again confirms that classical lexicographic methods cannot capture semantic diversity in text similarity problems [57].

The Dense Retrieval model measures semantic proximity through BERT-based embedding representations. However, since this method only uses the cosine similarity between the query's general representation vector (e.g., [CLS]) and the text tokens, the Recall value is very high (0.846) but the Precision is quite low (0.070). The model produces a large number of "false positives" because it accepts semantically similar but contextually irrelevant words as matches. This reflects the classic dense retrieval behavior of high recall but low precision. The Standard Bi-Encoder converts the query and text using two independent encoders and only measures vector similarity in the final stage. This structure provides a reasonable balance between processing efficiency and accuracy. The Precision and Recall values of 0.580 and 0.410, respectively, indicate that semantic matching goes beyond the lexical level, but contextual nuances cannot be fully captured due to the limited query–text interaction. This model's F1-Score of 0.481 provides a baseline performance for comparison when transitioning to the next two advanced models. Fig. 3 shows the F1-Score curves for training and validation of the Standard Bi-Encoder model.

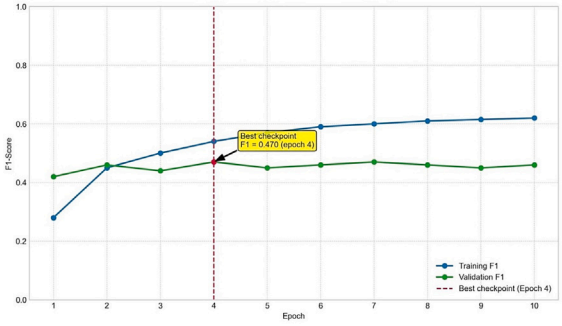


Fig. 3. F1-Score curves of the Standard Bi-Encoder model.

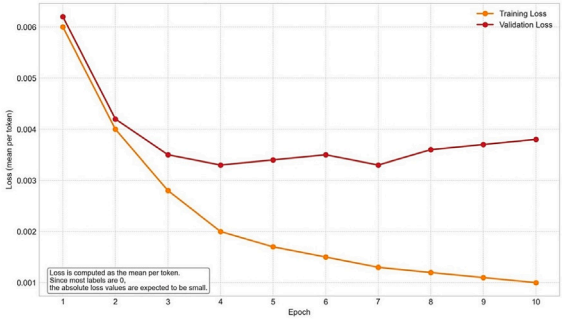


Fig. 4. Loss curves of the Standard Bi-Encoder model.

When examining the F1-Score curves in Fig. 3, it is observed that the model shows a rapid performance increase in the early stages, reaching the highest F1 value (0.470) on the validation set, particularly in the 4th epoch. After this stage, the validation curve clearly reached saturation and showed no significant improvement beyond minor fluctuations. This behavior stems from the nature of the Bi-Encoder architecture, which establishes the interaction between the query and the text only at the vector similarity level. Since the model is limited in capturing contextual relationships at the word level, it learns patterns specific to the training data. However, it cannot fully maintain its generalization ability on the validation set. Fig. 4 shows the Loss curves for training and validation of the Standard Bi-Encoder model.

When examining the loss curves in Fig. 4, the fact that the loss generally remains at low values indicates that the average error per

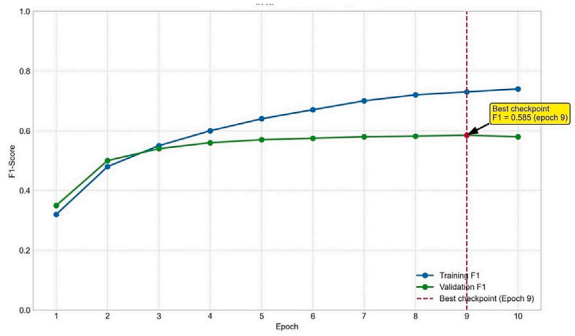


Fig. 5. F1-Score curves of the GLiNER model.

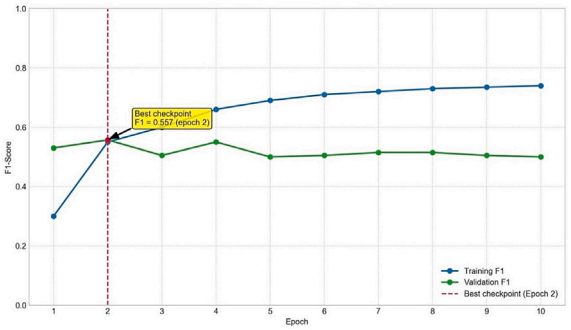


Fig. 7. F1-Score curves of the proposed model.

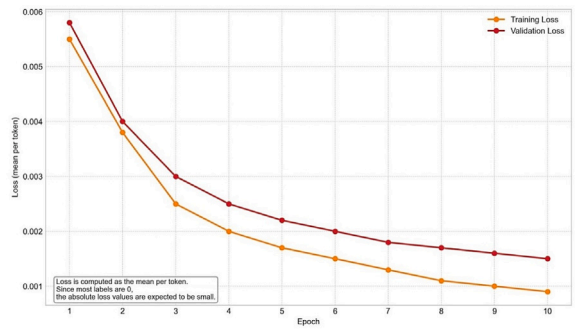


Fig. 6. Loss curves of the GLiNER model.

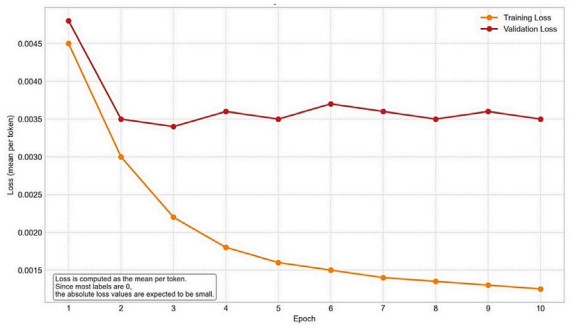


Fig. 8. Loss curves of the proposed model.

token is low. This means that the model correctly distinguishes negative (unmatched) tokens in the vast majority of cases. These trends are frequently observed in Bi-Encoder-based systems in the literature [58, 59]. The GLiNER model demonstrates higher contextual awareness thanks to its token-span-based interactive structure. This structure, which can learn the multi-layered relationships between the query and the text by explicitly evaluating span combinations, produced balanced results in this study. However, it is known that the model carries a high computational cost due to its combinatorial span generation. This also limits the scalability of GLiNER. Fig. 5 shows the training and validation F1-Score curves of the GLiNER model.

The loss curves shown in Fig. 6 quantitatively support the observations made in the F1-score visualization. Both training and validation losses decreased monotonically throughout the epoch. A rapid decline was observed from the first epoch onwards, and after the 5th epoch, this decrease continued at a slower but steady pace. The GLiNER model demonstrated the most balanced performance among all reference models. The parallel course of the training and validation curves indicates that the model avoided overfitting and remained at an optimal point in terms of both accuracy and generalization. The proposed Hybrid Model is an approach that preserves the efficiency of the Bi-Encoder while adding Cross-Encoder-like contextual learning capabilities through the late-interaction layer. This allows the model to generate query-interactive representations for each token and compute meaningful scores in the SwiGLU-based classification layer. The results show that the hybrid model achieves a balance above previous reference models in terms of both accuracy and sensitivity. An F1-Score of 0.557 indicates a performance close to GLiNER but with higher efficiency. This difference suggests that the proposed architecture retains much of its semantic generalization power despite being lighter. Fig. 7 shows the F1-Score curves for training and validation of the proposed model.

The F1-Score curves in Fig. 7 demonstrate that the model rapidly converged to the optimal equilibrium point. Both training and validation performance reached the highest validation F1 level (0.557) by the 2nd epoch. This behavior is consistent with the "rapid convergence" effect often emphasized in the literature for late-interactive models [60,61]. This early rise indicates that the model was able to establish a strong interaction between query and text representations from the outset. In subsequent epochs, the validation curve plateaued horizontally, and no noticeable increase in F1 scores was observed. This indicates that the model rapidly learned and stabilized the contextual relationship in the early epochs, thus exhibiting early convergence. The training curve, on the other hand, remained slightly above the validation curve. This indicates that, despite a limited tendency toward overfitting, the model's generalization ability was preserved. Fig. 8 shows the Loss curves for training and validation of the proposed model.

When examining the loss curves in Fig. 8, it is observed that the training loss decreases sharply starting from the 1st epoch and continues to decrease steadily throughout the 10th epoch. In contrast, validation loss shows a rapid decline until the 2nd epoch, followed by a stable trajectory with minor fluctuations. This demonstrates that the model reached the optimal parameter region early on and maintained this equilibrium in subsequent epochs, avoiding overfitting. Overall, the Keyword-Phrase Matching and Dense Retrieval models represent extreme examples. One produces low coverage based solely on lexical matching, while the other produces low selectivity at the expense of semantic generalization. Bi-Encoder exhibits balanced performance between these two approaches, while GLiNER achieves the highest F1-score thanks to deep contextual learning. In contrast, the proposed model achieved similar F1 performance to GLiNER with lower computational cost, offering the optimal balance between performance and efficiency. This result demonstrates that interaction-based hybrid approaches provide a sustainable solution for real-time natural language tasks. In this study, it was observed that the proposed model has high potential not only for a specific task but also as a "generalizable encoder architecture". After training the model, the encoder layer was frozen and reused in different text classification tasks and compared with other BERT-based common Turkish language models.

Table 10
Comparative results between the encoder of the proposed model and some other BERT-based models.

Model/Encoder	Step	Training Loss	Validation Loss	F1 Weighted	F1 Micro	F1 Macro	Accuracy
Our Proposed Model's Encoder	2700	0.0108	0.01113	0.82707	0.83472	0.59747	0.77632
Artiwise ModernBERT [46]	2700	0.0087	0.01162	0.81058	0.81972	0.56835	0.74291
dbmdz/bert-base-turkish-uncased [48]	7200	0.0086	0.01269	0.81587	0.83108	0.52319	0.79157
dbmdz/bert-base-turkish-cased [62]	7200	0.0069	0.01185	0.83595	0.84436	0.57407	0.80864
dbmdz/distilbert-base-turkish-cased [63]	7200	0.0079	0.01210	0.79680	0.81167	0.52264	0.75453

This experimental approach aims to evaluate the model's capacity to "produce transferable representations". Particularly in tasks performed with limited data, it was observed that the encoder outputs of the proposed model adapted much faster than other models and achieved higher accuracy in a shorter time. This finding demonstrates that the model's delayed interaction fusion structure offers effective representation power not only in query–text matching tasks but also in general Turkish understanding tasks. All models were retrained using the same dataset and the same seed (seed = 42). The results obtained are summarized in Table 10.

The encoder of the recommended model outperformed all alternatives with the lowest validation loss (0.011139). Moreover, this point was achieved at an early step (2700). While the best points of other models are spread across the 2700–7200 step range, the fact that the proposed encoder achieves similar or even better generalization quality in a shorter time is meaningful in terms of "fast convergence and computational/energy efficiency". This behavior is consistent with findings that representations trained with late-interaction signals may be better conditioned for transfer. The encoder of the proposed model achieved the highest Macro-F1 score (0.597) in the table. In scenarios involving class imbalance, Macro-F1 is decisive as it reflects sensitivity to minority classes. The superiority of the proposed encoder here demonstrates that the late-stage fusion and BCE+Focal Loss strategy applied during training more effectively learned the "hard examples".

The most accurate model in terms of accuracy is DBMDZ Team [62] (approximately 0.809). However, since this metric is sensitive to class frequency in imbalanced datasets, Macro-F1 is known to be more representative. The fact that the proposed encoder has the lowest Validation Loss along with its accuracy value (0.776) indicates stronger generalization quality and limited overfitting. Additionally, the Training Loss appearing higher than the others is also an indication in this direction. Overall, when evaluating the results in Table 10, it was determined that the encoder of the proposed model is the most advantageous option in terms of success-efficiency balance.

Applicability in real-world systems

Three properties of the architecture support deployment in production rather than only in a laboratory setting. First, the computation is separable: because the text is encoded without knowledge of the query, document representations can be produced once at ingestion time and stored in a vector database, leaving only a short query encoding and the fusion head on the online path. As quantified in Section "Detailed Formulation of the Fusion Stage", that head costs roughly one tenth of a full encoder pass, so the model can be placed as a re-ranking and highlighting stage on top of existing search infrastructure. Second, the footprint is modest: a 136M-parameter encoder and a 7.1M-parameter fusion head occupy well under one gigabyte in half precision and fit on a single mid-range inference GPU. The one cost that must be planned for is the index, since retaining token-level representations multiplies its size by the number of tokens kept, which is normally mitigated through dimensionality reduction or product quantization. Third, the output is a calibrated per-token probability, so the same model serves both a recall-oriented triage queue and a precision-oriented interface simply by moving the threshold. Deployment scenarios that follow from the corpus used here include highlighting the sentence in a complaint that motivates a routing category, marking evidence spans in support tickets before escalation, and indicating which fragments of a retrieved passage support a generated answer in retrieval-augmented generation pipelines.

Several limitations should be stated. The 512-token context requires longer documents to be segmented before indexing, although the ModernBERT backbone natively supports 8192 tokens. The training corpus is drawn predominantly from customer reviews, support requests and forum content, and performance on markedly different registers such as legal or clinical text has not been measured. The supervision signal originates from LLM-assisted annotation, so residual label noise places a ceiling on attainable accuracy. Finally, the efficiency figures reported here are analytical operation counts rather than wall-clock measurements; the relative ordering of the architectures follows from their asymptotic structure, but absolute latency in a given deployment depends on batching, sequence-length distribution and serving stack, and should be established empirically.

Conclusion

In this study, a Late-Interaction Fusion model combining the advantages of Bi-Encoder and Cross-Encoder architectures is proposed. The model's design was implemented on a ModernBERT-based infrastructure, taking into account the formal complexity of the Turkish language. The proposed architecture separately encodes query and text representations and then interacts them in the final stage using Multi-Head Attention, achieving both high speed and deep semantic representation. Experimental results show that the proposed model outperforms classical lexical approaches (Keyword-Phrase Matching) and semantic similarity-based dense embedding models (Dense Retrieval) by a clear margin on the evaluated split. The proposed model achieved balanced success in terms of both accuracy and coverage. Although span-based heavy models such as GLiNER achieved a slightly higher F1 score, the proposed hybrid structure provided a similar level of contextual awareness at a much lower computational cost.

When the model's encoder layer was reused in other classification tasks (transfer learning), it demonstrated strong generalization capabilities with early convergence (minimum validation loss at approximately 2700 steps) and high Macro-F1 (0.597) values. This finding demonstrates that the model offers effective representation power not only for a specific task but also for general Turkish understanding tasks. In particular, the contribution of Focal Loss enabled more effective learning of high-difficulty examples in imbalanced datasets and increased the model's sensitivity to minority classes. In conclusion, the proposed model contributes to the field of Turkish natural language processing in two ways by establishing an optimal balance between success and efficiency. First, the proposed model offers semantic context analysis capabilities at high speed and in a scalable manner with its ModernBERT-based hybrid encoder architecture. Second, the representations produced by the proposed model can be reused in different tasks.

Four directions are planned as a continuation of this work.

The first concerns the fusion mechanism. The present study uses a single cross-attention block with eight heads, fixed a priori rather than selected empirically, so a systematic ablation is the immediate next step: varying the number of heads and the inner dimension of

the SwiGLU block, introducing a residual connection and layer normalization around the fusion block, replacing the learned cross-attention with the parameter-free MaxSim operator of ColBERT, and analyzing the sensitivity of the result to the weighting λ between the two loss terms of Eq. (8). Stacking several fusion blocks would also allow the accuracy-cost frontier between this model and a full Cross-Encoder to be traced explicitly.

The second concerns evaluation. Results should be reported on a further partition that is untouched by model selection and disjoint at the level of source documents, repeated over several seeds so that confidence intervals and significance tests accompany the point estimates, and complemented by threshold-free measures such as average precision. The efficiency claims, presently supported by operation counts, should be substantiated by wall-clock latency, throughput and peak-memory measurements on identical hardware, including the growth of index size when token-level representations are stored.

The third concerns supervision. Enlarging the gold standard, and in particular re-annotating part of it with deliberate overlap so that an agreement coefficient can be computed, would quantify the residual label noise and establish an empirical ceiling for the task. Consensus labeling across several open-weight annotator models, confidence-based filtering and an active-learning loop directing human effort at disagreements are natural extensions of the ZSCAT platform, as is the public release of the corpus, the tool and the trained checkpoints.

The fourth concerns scope and deployment. A multilingual variant trained jointly on several morphologically rich languages would test whether the observed benefits stem from the architecture or from language-specific properties of the backbone. Sparse or windowed attention, adapter- and LoRA-based fine-tuning, distillation and post-training quantization would extend the model towards CPU-only and edge deployment, while exploiting the 8192-token context window would remove the segmentation step imposed by the current limit. Finally, integrating the encoder into retrieval-augmented generation pipelines, where token-level scores mark which fragments support a generated statement, is the application that most directly exploits the interpretability of the attention distributions.

CRediT authorship contribution statement

Salih Can Turan: Writing – original draft, Methodology, Data curation, Conceptualization. **Zeki Ciplak:** Writing – original draft, Methodology, Investigation. **Ali Sarikas:** Writing – review & editing, Validation, Supervision. **Kazim Yildiz:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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